
Olist Executive Intelligence E2E Validation

Executive Summary Executive Business Deliverable

Author:	System E2E Tester
Workspace:	DEFAULT
Date:	September 17, 2026
Recipient:	saad@example.com
Confidence Score:	100.0%

1. Executive Summary

- Overall financial performance indicates primary metric volume of \$13.59M with a +13.6% trajectory [SRC-KPI-1].
- Operating margins maintain stability at 98,666 [SRC-KPI-2], reflecting disciplined capital allocation.
- Machine learning anomaly scanning detected 5 variance spikes requiring operational monitoring [SRC-ANOM-1].
- The ARIMA(1,1,1) model projects a upward trend over the next 6 Months, with forecast revenue reaching approximately \$154,196.21 by 2019-03-01 [SRC-FC-1].

2. Key Performance Indicators

Total Revenue \$13.59M +13.6%	Total Orders 98,666 +11.1%	Average Order Value \$137.75 +4.2%	Freight & Delivery Value \$2.25M +8.7%
--	---	---	---

3. Strategic Business Insights

Insight Title & Analysis	Severity	Supporting Metric
Executive Summary Volume Performance Primary metric generated \$13.59M, representing a +13.6% shift compared to prior baseline.	Low	\$13.59M
Statistical Anomaly Variance Isolation Forest identified 5 notable outlier points, with peak variance of +10.4 Std Dev on 2017-11-24.	High	+10.4 Std Dev
Customer Cohort Distribution Top segment 'High-Value Enterprise & VIP' accounts for 10.7% of records with average transaction value of \$546.22.	Medium	10.7%

4. Performance Snapshot



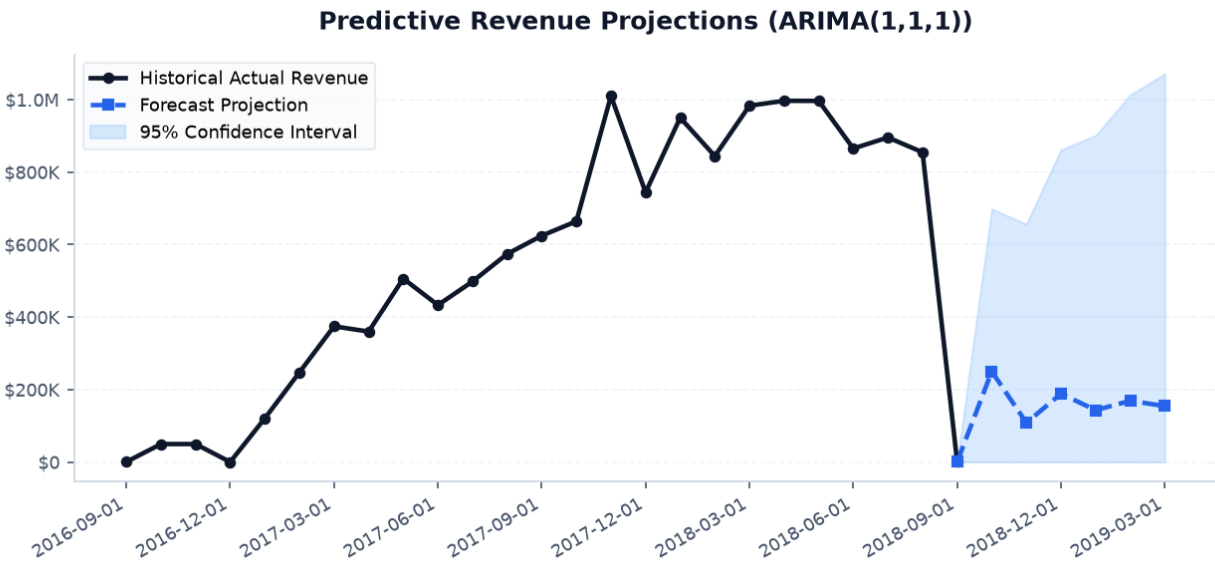
5. Anomaly Detection & Statistical Outliers

Metric Outlier	Affected Date	Severity	Deviation	Baseline
Daily Revenue Spike	2017-11-24	High	+10.4 Std Dev	\$22,064.36
Daily Revenue Spike	2017-11-25	High	+3.1 Std Dev	\$22,064.36
Daily Revenue Spike	2018-08-06	Medium	+2.7 Std Dev	\$22,064.36
Daily Revenue Spike	2018-05-16	Medium	+2.7 Std Dev	\$22,064.36
Daily Revenue Spike	2018-05-07	Medium	+2.5 Std Dev	\$22,064.36

6. Predictive Forecasting Projections

Forecast Horizon: 6 Months | Model: ARIMA(1,1,1) | Trajectory: Upward

Revenue is forecast to upward by 116300.2% over the next 6 monthly.



Forecast Period	Predicted Revenue	Lower Bound (95% CI)	Upper Bound (95% CI)
2018-10-01	\$249,836.73	\$0.00	\$698,408.46
2018-11-01	\$108,264.79	\$0.00	\$656,407.89
2018-12-01	\$188,534.22	\$0.00	\$861,693.80

2019-01-01	\$143,022.50	\$0.00	\$900,729.72
2019-02-01	\$168,827.05	\$0.00	\$1,012,996.08
2019-03-01	\$154,196.21	\$0.00	\$1,071,278.94

Evaluation Metrics: MAE: \$112,161.08 | RMSE: \$205,809.50 | MAPE: 43700.50%

Source: Forecasting Service [SRC-FC-1]

7. Customer Cohort & Segmentation

Segment Name	Size	Share	Avg Spend	Risk Rating
High-Value Enterprise & VIP	1,071	10.7%	\$546.22	Low
Core Marketplace Buyers	4,155	41.5%	\$138.54	Medium
Occasional & At-Risk Cohort	4,774	47.7%	\$44.87	High

8. Strategic Recommendations

Actionable Recommendation	Priority	Owner / Team
Investigate the top outlier transactions identified on 2017-11-24 and reconcile with gateway logs.	High	Finance & Risk Operations
Initiate proactive quarterly account reviews for the High-Value Enterprise & VIP customer cohort.	High	Customer Success & Enterprise Sales
Align Q4 operational inventory and staffing capacity with 30-day upward forecast trajectory.	Medium	Supply Chain & Operations

9. Evidence Grounding & Source Citations

[SRC-SQL-1] **SQL Analytics** (DuckDB Analytical Engine): DuckDB verified net transaction volume is 98,666 orders totaling \$13,591,643.70 with average order value \$137.75.

[SRC-KPI-1] **Dashboard KPI** (Dashboard KPI Engine): Total Revenue achieved \$13.59M (+13.6% period-over-period) with 98,666 fulfilled orders.

[SRC-FC-1] **Forecasting** (Forecasting Service (ARIMA(1,1,1))): ARIMA(1,1,1) forecast of monthly revenue for the next 6 months based on order history from 2016-09-01 to 2018-09-01. Final forecast period: 2019-03-01. Predicted revenue: \$154,196.21.

[SRC-SEG-1] **Segmentation** (K-Means Segmentation Engine): Primary segment 'High-Value Enterprise & VIP' accounts for 10.7% of customers with average spend of \$546.22.

[SRC-ANOM-1] **Anomaly Detection** (Statistical Outlier Detector): Detected 5 significant outlier events, largest on 2017-11-24 (+10.4 Std Dev).

[SRC-RAG-DICT] **Knowledge Base** (olist_business_dictionary.md): Verified Business Definitions: Dataset: Olist Brazilian E-Commerce; orders;; - order_id; - customer_id; - order_status; - order_purchase_timestamp